Junhyung Park

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Education

Nov.2019 – Jul.2024 **PhD in Machine Learning**, *Empirical Inference Department, Max-Planck-Institute for Intelligent Systems*, Tübingen, Germany, supervised by **Krikamol Muandet** and **Bernhard Schölkopf**.
Thesis: "A Measure-Theoretic Axiomatisation of Causality and Kernel Regression".

Sep.2017 – Aug.2019 **MSc in Statistics**, *Seminar für Statistik, Dept. of Mathematics, ETH Zürich*, Switzerland
Thesis: "Kernel Measures of Conditional Dependence", supervised by **Sara van de Geer**
Brownian Motion and Stochastic Calculus, Fundamentals of Mathematical Statistics, Computational Statistics, Introduction to Machine Learning, etc.

Oct.2015 – Jun.2016 **MMath in Mathematics**, *Trinity College, University of Cambridge*, Cambridge, UK
Commutative Algebra, Functional Analysis, Differential Geometry, Representation Theory, etc.

Oct.2012 – Jun.2015 **BA in Mathematics**, *Trinity College, University of Cambridge*, Cambridge, UK
Galois Theory, Linear Analysis, Differential Geometry, Number Fields, Complex Analysis, etc.

Work Experience

Oct.2026 – **DFG Walter Benjamin Postdoctoral Researcher**, *Technical University of Munich*, Germany
Supervised by **Mathias Drton**, guest contract at Helmholtz AI, Munich with **Niki Kilbertus**

Oct.2024 – Sep.2026 **Postdoctoral Researcher**, *ETH Zürich*, Switzerland
Supervised by **Fanny Yang**

Sep.2024 – Dec.2024 **Visiting Scholar**, *Simons Institute, UC Berkeley*, California, US
Programme: **Modern Paradigms in Generalization**

Aug.2023 – Dec 2023 **Applied Scientist Intern**, *Amazon Web Services*, Santa Clara, California, US
Supervised by **Shiva Kasiviswanathan** and **Patrick Blöbaum**

Sep.2016 – Jun.2017 **Junior Researcher in Statistics**, *Caleb ABC*, Seoul, Korea
Developed data envelopment analysis and logistic regression features on B-Box, a statistics software

Grants

Oct.2026 – Sep.2028 **DFG (German Research Foundation) Walter Benjamin Position**

Declined due to duplication **Humboldt Research Fellowship for Postdocs**

Tutorial

UAI 2026 **Park, J.** Causal spaces: a mathematical axiomatisation of causality

Books

In preparation, 2026 **Park, J.** A Primer on Causal Spaces

Publications

Submitted, 2026 **Park, J., Yang, F. and Icard, T.** Counterfactual spaces

CLeaR 2026 Oral **Park, J. and Zhou, Y.** A fine-grained look at causal effects in causal spaces

AISTATS 2026 **Näf, J., Park, J. and Susmann, H.** Causal-DRF: Conditional kernel treatment effect estimation using distributional random forest

NeurIPS 2025 Spotlight **Wegel, T., So, G., Park, J. and Yang, F.** On the sample complexity of semi-supervised multi-objective learning

UAI 2024 **Buchholz, S.*, Park, J.* and Schölkopf, B.** Products, abstractions and inclusions of causal spaces

PCIC 2024 **Dhanakshirur, M., Laumann, F., Park, J. and Barahona, M.** A continuous structural intervention distance to compare causal graphs

NeurIPS 2023 Oral **Park, J., Buchholz, S., Schölkopf, B. and Muandet, K.** A measure-theoretic axiomatisation of causality

Entropy 2023 **Laumann, F., von Kügelgen, J., Park, J., Schölkopf, B. and Barahona, M.** Kernel-based independence tests for causal structure learning on functional data

ALT 2023 **Park, J. and Muandet, K.** Towards empirical process theory for vector-valued functions: metric entropy of smooth function classes

ICML 2021 **Park, J., Shalit, U., Schölkopf, B. and Muandet, K.** Conditional distributional treatment effect with kernel conditional mean embeddings and U-statistic regression

arXiv 2020 **Park, J. and Muandet, K.** Regularised least-squares regression with infinite dimensional output space

NeurIPS 2020 **Park, J. and Muandet, K.** A measure-theoretic approach to kernel conditional mean embeddings

Scientific Meetings Organised

ICML 2026 Workshop Philosophy Meets Machine Learning: What Counts as Trustworthy?

MLSS 2020 Machine Learning Summer School 2020, Tübingen (virtual due to Covid19)

Presentations

Nov 2026 Technical University of Hamburg, **Causal Spaces: A Mathematical Axiomatisation of Causality**

Nov 2026 University of Potsdam, **Causal Spaces: A Mathematical Axiomatisation of Causality**

Aug 2026 Technical University of Munich, **Causal Spaces: A Mathematical Axiomatisation of Causality**

Jun 2026 University Paris-Saclay, **Causal Spaces: A Mathematical Axiomatisation of Causality**

Jun 2026 University of Manchester, **Causal Spaces: A Mathematical Axiomatisation of Causality**

Jun 2026 University of Oxford, **Causal Spaces: A Mathematical Axiomatisation of Causality**

Jun 2026 University College London, **Targeted Interventions in Causal Spaces**

Jun 2026 PreMeDICaL Inria-Inserm, Montpellier, **Targeted Interventions in Causal Spaces**

Jun 2026 Amsterdam Causality Meeting, **Causal Spaces: A Mathematical Axiomatisation of Causality**

May 2026 Yonsei University, **Causal Spaces: A Mathematical Axiomatisation of Causality**

Nov 2025 University College London, **Causal and Counterfactual Spaces**

Oct 2025 Technical University of Munich, **Causal and Counterfactual Spaces**

Oct 2025 Symposium on Mathematical Foundations of Trustworthy Learning, **Causal and Counterfactual Spaces**

Oct 2025 ETH Zürich, **Counterfactual Spaces**

Jul 2025 MPI for Intelligent Systems, Tübingen, **Causal and Counterfactual Spaces**

Jul 2025 Technical University of Munich, **A Measure-Theoretic Axiomatisation of Causality**

Jun 2025 German Research Center for Artificial Intelligence, **A Measure-Theoretic Axiomatisation of Causality**

Apr 2025 Seoul National University, **A Measure-Theoretic Axiomatisation of Causality**

Jan 2025 Korea University, **A Measure-Theoretic Axiomatisation of Causality**

Jan 2025 Korean Institute for Advanced Study, **A Measure-Theoretic Axiomatisation of Causality**

Nov 2024 University of Pisa, **A Measure-Theoretic Axiomatisation of Causality**

Oct 2024 Seminar for Statistics, ETH Zürich, **A Measure-Theoretic Axiomatisation of Causality**

Aug 2024 University of Copenhagen, **A Measure-Theoretic Axiomatisation of Causality**

Feb 2024 ETH Zürich, **A Measure-Theoretic Axiomatisation of Causality**

Nov 2023 Leipzig University, **A Measure-Theoretic Axiomatisation of Causality**

Nov 2023 Amazon, **A Measure-Theoretic Axiomatisation of Causality**

Oct 2023 Stanford University, USA, **A Measure-Theoretic Axiomatisation of Causality**

Jul 2023 European Meeting of Statisticians, Warsaw, Poland, **Towards a Measure-Theoretic Axiomatisation of Causality**

May 2023 Colloquium on "Fundamental Challenges in Causality", Grenoble, France, **Towards a Measure-Theoretic Axiomatisation of Causality**

Apr 2023 Colloquium on "When Causal Inference meets Statistical Analysis", Paris, France, **Towards a Measure-Theoretic Axiomatisation of Causality**

Apr 2023 Workshop on Causal Representation Learning, Tübingen, Germany, **Towards a Measure-Theoretic Axiomatisation of Causality**

Mar 2023 CISPA – Helmholtz Center for Information Security, Saarbrücken, Germany, **Towards a Measure-Theoretic Axiomatisation of Causality**

Sep 2022 ELISE Theory workshop on machine learning fundamentals, Antibes, France, **Kernel Conditional Mean Embeddings and Empirical Process Theory for Vector-Valued Functions**

Jul 2022	Saint Flour Probability Summer School, Saint Flour, France, **Empirical Process Theory for Vector-Valued Functions**
Mar 2022	Cornell University, USA, **Distributional Treatment Effects with Kernels**
Jul 2020	Petnica Summer Institute Machine Learning (Microsoft, Belgrade), **Kernel Methods in Machine Learning**

Supervised Students

Winter 2026/27	Felix Reich, Junhong Chong, Gereon Mömkes (Bachelor's theses at TUM)
Summer 2026	Ciaran Wehrli, Dimitris Skikos, Leo An (Bachelor's projects at ETH Zürich)
Autumn 2025, Summer 2026	Yuqing Zhou, semester project & Master's thesis at ETH Zürich

Teaching Experience

Winter 2026/27	Fundamentals of Mathematical Statistics (TUM, substitute lecturer)
Spring 2026	Introduction to machine learning (ETH Zürich, TA)
Autumn 2025	Advanced topics in machine learning, Guarantees in machine learning (ETH Zürich, TA)
Spring 2025	Introduction to machine learning (ETH Zürich, TA)

Reviews

NeurIPS	2021, 2022, 2023, 2024, 2025
ICML	2022, 2023, 2024, 2025
JMLR	2025, 2026
AoS	2021, 2023
ICLR	2022
AISTATS	2022 (Top 10% of reviewers), 2023, 2026
UAI	2025, 2026
Biometrika	2023
Analysis and Applications	2023
CDC	2022
TPAMI	2024
TMLR	2025, 2026
IMAIAI	2026

Technical & Language Skills

Programming	Python, R, MATLAB
Software	LaTeX
Languages	Korean (Mother Tongue), English (Bilingual), French (DALF C1, May 2026), German (Elementary)

Additional Activities

Football	Member of Trinity College 1st Football Team (2012–2016) with captaincy (2013–2014). Member of TV Derendingen (2021–2024)
Music	Piano, grade 8, Associated Board of the Royal Schools of Music, UK